Math 121: Commutative Algebra

Prishita Dharampal

Worksheet 7.1: Integral Extension Pt. I

Throughout this course, "ring" means *commutative* ring with unity and all ring homomorphisms preserve unity. In the previous worksheets we have repeatedly moved between rings, modules, quotients, localization, and spectra. The goal of this worksheet is to introduce a language that lets us treat many ring maps as geometric families: the language of *R-algebras*. Once this language is in place, we will distinguish two finiteness notions that sound similar but behave very differently: finite generation as an algebra and finite generation as a module.

The main bridge between these two notions is integrality. Integral elements are elements of an extension ring that satisfy monic polynomial equations over the base. The monic condition is what lets one convert algebraic equations into module-theoretic finiteness. This is one of the basic mechanisms behind many finiteness theorems in commutative algebra and algebraic geometry.

Let R be a ring. An *R-algebra* is a ring S together with a specified ring homomorphism $\alpha : R \longrightarrow S$. Note we do not require that α be injective, however, the case when α is an inclusion, i.e., $\mathbb{K}$ into $\mathbb{K}[x]$, is generally the motivating example. We often think of an R-algebra structure on a ring S as giving us a way to multiply elements of S by elements of R, i.e., $r \cdot s := \alpha(r)s$. Thus every R-algebra S is naturally an R-module. The key difference between an R-module M and an R-algebra S is that for module we may only multiple elements of M by elements of R, but for an R-algebra we every element of S may be multiplied together. The key analogy might be: R-modules are to vector spaces, as R-algebras are to rings.

An R-algebra homomorphism $\phi : S \to T$ is a ring homomorphism compatible with the structure maps from R, meaning that the diagram

$$\begin{tikzcd} S \arrow[rr, "\phi"] \arrow[dr, "\alpha_S"'] & & T \arrow[dl, "\alpha_T"] \\ & T & \end{tikzcd}$$

commutes. Equivalently, $\phi(\alpha_S(r)) = \alpha_T(r)$ for every $r \in R$. An *R-subalgebra* of S is a subring $S' \subset S$, which contains $\alpha(R)$. Given any set $\mathcal{S} \subset S$ the *R-algebra generated by* $\mathcal{S}$ is the smallest R-subalgebra of S containing $\mathcal{S}$. We often write $R[\mathcal{S}]$ to denote R-subalgebra generated by $\mathcal{S}$, because it can explicitly be written down as the set of polynomials in the elements in $\mathcal{S}$ with coefficients in R.

Problem 1. First Encounters with R-algebras.
Let $\alpha : R \to S$ be an R-algebra.

1. Verify that the rule $r \cdot s = \alpha(r)s$ makes S into an R-module.
2. Give an example where α is not injective. In your example, identify a nonzero element of R that acts as zero on S.
3. Show that every quotient R/I is naturally an R-algebra.
4. Show that if S and T are R-algebras and $\phi : S \to T$ is an R-algebra homomorphism, then ϕ is R-linear for the underlying R-module structures.
5. Let $\mathbb{K} \subseteq \mathbb{E}$ be a field extension. Explain why $\mathbb{E}$ is a $\mathbb{K}$-algebra. What is the underlying $\mathbb{K}$-module structure?

Solution.

1. We verify the axioms. For all $r, p \in R, s, t \in S$ we have:
   (a) $r(s+t) = \alpha(r)(s+t) = \alpha(r)s + \alpha(r)t = rs + rt$
   (b) $(r+p)t = \alpha(r+p)t = \alpha(r)t + \alpha(p)t = rt + pt$
   (c) $(rp)t = \alpha(rp)t = \alpha(r)\alpha(p)t = \alpha(r)(\alpha(p)t) = r(pt)$

   Thus the rule $r \cdot s = \alpha(r)s$ makes S into a R-module.
2. $\alpha : \mathbb{Z} \to \mathbb{Z}/2\mathbb{Z}$.
   2 is a non-zero element in $\mathbb{Z}$ but acts as zero on $\mathbb{Z}/2\mathbb{Z}$.
3. The natural projection $\pi : R \to R/I$ is a ring homomorphism, and the property we proved in part (1) works with substituting $\alpha = \pi$.
4. Let $\alpha : R \to S$ and $\beta : R \to T$. Then, for $r, s \in R$,
$$\phi(r \cdot s) = \phi(\alpha(r)s) = \phi(\alpha(r))\phi(s) = \beta(r)\phi(s) = r\phi(s)$$
   Hence, ϕ is R-linear.
5. $\mathbb{E}$ is a $\mathbb{K}$-algebra with the inclusion map $\iota : \mathbb{K} \to \mathbb{E}$, $\iota(k) = k$ as the ring homomorphism. Moreover multiplication in $\mathbb{E}$ is $\mathbb{K}$-linear and,
$$k \cdot e = \iota(k)e = ke$$

Problem 2. Polynomial Rings and Generation of R-algebras.
Let S be an R-algebra and let $s_1, \ldots, s_n \in S$.

1. Construct an R-algebra homomorphism
$$\mathrm{ev}_{s_1,\ldots,s_n} : R[x_1, \ldots, x_n] \longrightarrow S, \qquad f(x_1, \ldots, x_n) \longmapsto f(s_1, \ldots, s_n).$$
2. Show that the image of this map is the smallest R-subalgebra of S containing $s_1, \ldots, s_n$. We denote this image by $R[s_1, \ldots, s_n]$.
3. Prove that $S = R[s_1, \ldots, s_n]$ if and only if $\mathrm{ev}_{s_1,\ldots,s_n}$ is surjective.
4. Deduce that S is generated as an R-algebra by n elements if and only if $S \cong R[x_1, \ldots, x_n]/I$ as an R-algebra for some ideal $I \subset R[x_1, \ldots, x_n]$.

Solution.

1. We define $\mathrm{ev}_{s_1,\ldots,s_n} : R[x_1, \ldots, x_n] \longrightarrow S$ by,
$$\mathrm{ev}_{s_1,\ldots,s_n}(f(x_1, \ldots, x_n)) = \mathrm{ev}_{s_1,\ldots,s_n}\left(\sum_i r_i x_i\right) = f(s_1, \ldots, s_n) = \sum_i r_i(s_i) \in S$$
This is a homomorphism since both
$$(f+g)(s_1, \ldots, s_n) = \sum_i (r_i + r_i')s_i = \sum_i r_i s_i + \sum_i r_i' s_i = f(s_1, \ldots, s_n) + g(s_1, \ldots, s_n)$$
and,
$$(fg)(s_1, \ldots, s_n) = \sum_i (r_i \cdot r_i')s_i = f(s_1, \ldots, s_n) \cdot g(s_1, \ldots, s_n)$$
hold.

2. We know that $\mathrm{im}(\mathrm{ev}_{s_1,\ldots,s_n}) = R[s_1, \ldots, s_n]$. Let $T \subseteq S$ be a R-subalgebra with $s_1, \ldots, s_n \in T$. Note that a general element of $R[s_1, \ldots, s_n]$ has the form
$$f(s_1, \ldots, s_n) = \sum_i r_i s_1^{i_1} \cdots s_n^{i_n}$$
But since T is a R-subalgebra the sum $\sum_i r_i s_1^{i_1} \cdots s_n^{i_n} \in T$ since T is closed under addition and multiplication and $r_i \in T$ for all $r_i \in R$. Thus, $R[s_1, \ldots, s_n] \subseteq T$. Since $R[s_1, \ldots, s_n]$ is contained in every R-subalgebra of S containing $s_1, \ldots, s_n$ and is itself such a subalgebra it must be the smallest one.

3. We know that $\operatorname{im}(\operatorname{ev}_{s_1,\ldots,s_n}) = R[s_1,\ldots,s_n] \subseteq S$. If $S = R[s_1,\ldots,s_n]$ then

$$S = \operatorname{im}(\operatorname{ev}_{s_1,\ldots,s_n})$$

and $\operatorname{ev}_{s_1,\ldots,s_n}$ are surjective.

4. $(\implies)$

Assume S is a R-algebra generated by $s_1,\ldots,s_n \in S$. Then we have $S = R[x_1,\ldots,x_n] = \operatorname{im}(\operatorname{ev}_{s_1,\ldots,s_n})$ by part 3. Then by the first isomorphism theorem,

$$S = \operatorname{im}(\operatorname{ev}_{s_1,\ldots,s_n}) = R[x_1,\ldots,x_n]/I$$

for some $I \subseteq R[x_1,\ldots,x_n]$.

$(\impliedby)$

Assume $S \cong R[x_1,\ldots,x_n]/I$. Then there exists a isomorphism $\phi : R[x_1,\ldots,x_n]/I \to S$, where $s_i = \phi(x_i + I)$ for $1 \leq i \leq n$. Then every element of S can be written as

$$\phi(f+I) = \phi\left(\sum_i r_i x_1^{i_1}\cdots x_n^{i_n} + I\right) = r_i s_1^{i_1}\cdots s_n^{i_n} = f(s_1,\ldots,s_n)$$

so every element of S is a polynomial expression in $s_1,\ldots,s_n$ with coefficients in R. Hence $S = R[s_1,\ldots,s_n]$ is an R-algebra generated by n elements.

Problem 3. Examples of R-algebras.
In each case, identify a set of algebra generators over the indicated base ring. Try to choose a minimal-looking set, though you do not need to prove minimality.

1. $R[x]/\langle x^2\rangle$ over R.
2. $R[x,y]/\langle xy\rangle$ over R.
3. R/I over R.
4. $\mathbb{K}[t,t^{-1}]$ over $\mathbb{K}$.
5. $\mathbb{Z}[i]$ over $\mathbb{Z}$.
6. $\mathbb{K}[x,y]/\langle y^2-x^3\rangle$ over $\mathbb{K}[x]$.
7. $\mathbb{K}(t)[[t]]$ over the $\mathbb{K}(t)$.
8. $\mathbb{Z}[\sqrt{n}; n\in\mathbb{Z}_{\geq 1}]$ over $\mathbb{Z}$.

Solution.

1. $\{\overline{x}\}$
2. $\{\overline{x},\overline{y}\}$
3. $\{\}$
4. $\{t,t^{-1}\}$
5. $\{i\}$
6. $\{\overline{y}\}$
7. **TODO:**
8. $\{\sqrt{p}\mid p\text{ prime}\}$

As discussed above an R-algebra S is also always an R-module with the same R-multiplication. This leads to two different finiteness conditions depending on whether we wish to view S as an R-module or as an R-algebra. Let S be an R-algebra.

Definition 0.1. *We say that S is* algebra-finite over R *if S is finitely generated s as an R-algebra.*

Definition 0.2. *We say that S is* module-finite over R *if S is finitely generated as an R-module.*

The condition S be algebra-finite over R is sometimes called being *finite type* over R, i.e. we would say S is finite type over R if and only if S is algebra-finite over R. Similarly, many authors will use the phrase S is *finite* over R or S is a *finite R-algebra* to mean S if a R-algebra that is module-finite,. In this worksheet we will attempt to restrict to using *algebra-finite* and *module-finite* to hopefully minimize confusion.

The key point is that module-finite is much stronger than algebra-finite. A module-finite algebra is controlled by finitely many R-linear combinations. An algebra-finite algebra is controlled by finitely many elements and all polynomial expressions in them.

Problem 4. **Module-Finite Implies Algebra-Finite.**
Let S be an R-algebra and suppose that S is generated as an R-module by $m_1, \ldots, m_n \in S$.

1. Show that the R-subalgebra $R[m_1, \ldots, m_n]$ contains every R-linear combination of $m_1, \ldots, m_n$.
2. Deduce that $S = R[m_1, \ldots, m_n]$.
3. Conclude that every module-finite R-algebra is algebra-finite over R.

Solution.

1. An R-linear combination is of the form $r_1 m_1 + \cdots r_n m_n$ for all $r_i \in R$. But this is the polynomial $r_1 x_1 + \cdots r_n x_n$ evaluated at $m_1, \ldots, m_n$ which is an element of $\mathrm{im}(ev_{m_1,\ldots,m_n}) = R[m_1, \ldots, m_n]$.
2. The containment $S \supseteq R[m_1, \ldots, m_n]$ is obvious because the latter is a subalgebra of S and part 1 gives us the reverse containment, hence we have $S = R[m_1, \ldots, m_n]$.
3. Thus every module-finite R-algebra generated by n elements can be written in the form $R[m_1, \ldots, m_n]$ (by part 2) and thus is algebra-finite over R.

Problem 5. Algebra-Finite Need Not Imply Module-Finite.
Assume R is nonzero.

1. Show that $R[x]$ is algebra-finite over R.
2. Suppose $p_1, \ldots, p_n \in R[x]$. Let d be the maximum of their degrees. Show that every R-linear combination of the p_i has degree at most d, unless it is zero.
3. Deduce that $R[x]$ is not module-finite over R. Hint: Is x^{d+1} in the R-span of $p_1, \ldots, p_n$?
4. Specialize to $R = \mathbb{K}$ a field and restate the result in the language of vector spaces.

Solution.

1. $R[x]$ is generated by $\{x\}$ as an algebra and hence is algebra-finite.
2. Every R-linear combination of the p_i is of the form

$$r_1 p_1 + \cdots + r_n p_n, \quad r_i \in R$$

   since $\deg(r_i) = 0$ the degree of the equation is the maximum of the degrees of the p_i which is d.
3. Assume $R[x]$ is module-finite over R. Then there exists a finite set of generating elements $p_1, \ldots, p_n \in R[x]$. Let d be the maximum of their degrees. Now, $x^{d+1} \in R[x]$ with degree $d+1$ but by part 2 we know that any R linear combination of the p_i has degree at most d — a contradition. Hence, $R[x]$ is not module finite over R.
4. If $R = \mathbb{K}$ then $R[x]$ is an infinite dimensional vector space.

Problem 6. **Permanence of Algebra-Finiteness.**
The goal of this exercise is to explore how being algebra-finite plays with many natural algebraic operations: quotienting, tensoring, localizing, and composing the underlying morphisms giving the algebra structure. Consider maps of rings:

$$R \xrightarrow{\alpha} S \xrightarrow{\beta} T \ .$$

The map α gives S an R-algebra structure. On the other hand T has two different algebra structures: we may think of T as an S-algebra via the map β or think of T as an R-algebra via $\beta \circ \alpha$.

1. If S is algebra-finite over R and $J \subseteq S$ is an ideal, prove that S/J is algebra-finite over R.
2. If S is algebra-finite over R and T is algebra-finite over S, prove that T is algebra-finite over R.
3. Give an example where α and β are injective and T is algebra-finite over R, but S is not algebra-finite over R. Hint: Let $R = \mathbb{K}$, and $T = \mathbb{K}[x, y]$, what happens if $S = \mathbb{K}[x, xy, xy^2, xy^3, \ldots]$?
4. If S is algebra-finite over R, and $R \to T$ is a ring map, prove that $T \otimes_R S$ is algebra-finite over T. What are algebra generators?
5. If $U \subset R$ is multiplicatively closed and S is algebra-finite over R, prove that $U^{-1}S$ is algebra-finite over $U^{-1}R$.

Solution.

1. Let $S = R[s_1, \ldots, s_n]$. Then the quotinent map $\pi : S \to S/J$ is a surjective homomorphism with $\text{im}(\pi) = R[\overline{s_1}, \ldots, \overline{s_n}]$, where $\pi(s_i) = \overline{s_i}$. Hence S/J is algebra-finite over R.
2. Let $S = R[s_1, \ldots, s_n]$, and $T = S[t_1, \ldots, t_m]$. Then every element of T is a polynomial in $t_1, \ldots, t_m$ with coefficients in S, and each coefficient is a polynomial in $s_1, \ldots, s_n$ with coefficients in R. Hence every element of T is a polynomials expression in $\beta(s_1), \ldots, \beta(s_n), t_1, \ldots, t_m$ with coefficients in R, so

$$T = R[\beta(s_1), \ldots, \beta(s_n), t_1, \ldots, t_m]$$

which is algebra finite over R.
3. We let $R = \mathbb{K}$, $T = \mathbb{K}[x, y]$ and $S = R[x, xy, xy^2, xy^3, \ldots]$.

4. Let $S = R[s_1, \ldots, s_n]$. Every element of $T \otimes_R S$ is a finite sum of simple tensors $t \otimes s$. Since $s \in S$, we can write $s = \sum_i r_i s_1^{i_1} \ldots s_n^{i_n}$ for $r_i \in R$. Then,

$$\begin{aligned} t \otimes s &= t \otimes \sum_i r_i s_1^{i_1} \cdots s_n^{i_n} \\ &= t \sum_i r_i \otimes \sum_i s_1^{i_1} \cdots s_n^{i_n} \\ &= \sum_i tr_i \otimes s_1^{i_1} \cdots s_n^{i_n} \\ &= \sum_i tr_i (1 \otimes s_1)^{i_1} \cdots (1 \otimes s_n)^{i_n} \end{aligned}$$

Which is a polynomial in $(1 \otimes s_1), \ldots, (1 \otimes s_n)$ with coefficients in T and thus, these are the algebraic generators of $T \otimes_R S$ and $T \otimes_R S = T[1 \otimes s_1, \ldots, 1 \otimes s_n]$.

5. Let $S = R[s_1, \ldots, s_n]$. Note that the map $\eta : S \to U^{-1}S$ is a R-algebra homomorphism, and hence a $U^{-1}R$ algebra homomorphism (by extending scalars). Every element of $U^{-1}S$ is of the form $\frac{s}{u} = \frac{1}{u} \cdot \frac{s}{1}$ where $\frac{1}{u} \in U^{-1}R$.

   Since every element $s \in S$ is of the form $s = \sum_i r_i s_1^{i_1} \cdots s_n^{i_n}$ we get

$$\frac{s}{u} = \frac{1}{u} \sum_i \frac{r_i}{1} \left(\frac{s_1}{1}\right)^{i_1} \cdots \left(\frac{s_n}{1}\right)^{i_n}$$

   which is a polynomial in $\frac{s_1}{1}, \ldots, \frac{s_n}{1}$. So, $U^{-1}S = U^{-1}R[s_1, \ldots, s_n]$ which is algebra-finite over R.

Problem 7. Permanence of Module-Finiteness.
The goal of this exercise is to explore how being module-finite plays with many natural algebraic operations: quotienting, tensoring, localizing, and composing the underlying morphisms giving the algebra structure. Consider maps of rings:

$$R \xrightarrow{\alpha} S \xrightarrow{\beta} T \ .$$

The map α gives S an R-module structure. On the other hand T has two different module structures: we may think of T as an S-module via the map β or think of T as an R-module via $\beta \circ \alpha$.

1. If S is module-finite over R and $J \subseteq S$ is an ideal, prove that S/J is module-finite over R.
2. If S is module-finite over R and T is module-finite over S, prove that T is module-finite over R.
3. If S is module-finite over R, and $R \to T$ is a ring map, prove that $T \otimes_R S$ is module-finite over T.
4. If $U \subset R$ is multiplicatively closed and S is module-finite over R, prove that $U^{-1}S$ is module-finite over $U^{-1}R$.

Solution.

1. Let $S = R[m_1, \ldots, m_n]$. Then, every element in $s \in S$ can be written as $s = \sum_i r_i m_i$ for $r_i \in R$, $m_i \in S$. Then, every element of S/J can be written as $\bar{s} = \sum_i \overline{r_i}\, \overline{m_i} = \sum_i r_i \overline{m_i}$. Thus S/J is spanned by the $\overline{m_i}$ as a R-module and hence is module-finite.
2. Let $T = S[n_1, \ldots, n_m]$ and $S = R[m_1, \ldots, m_n]$. Then every element in T can be written as $T = \sum_i s_i n_i$ with $s_i \in S$, and every s_i can be written as $\sum_j r_j m_j$. Then every element in T can be written as
$$t = \sum_i \left(\sum_j r_{ij} m_j \right) n_i = \sum_{i,j} r_{ij} (m_j n_i)$$
Thus T is spanned by the $m_j n_i$ as a R-module and hence is module-finite.
3. Let $S = R[m_1, \ldots, m_n]$. Every element of $T \otimes_R S$ is a finite sum of simple tensors

$t \otimes s$. Since $s \in S$, we can write $s = \sum_i r_i m_i$ for $r_i \in R$. Then,

$$\begin{aligned} t \otimes s &= t \otimes \sum_i r_i m_i \\ &= t \sum_i r_i \otimes \sum_i m_i \\ &= \sum_i t r_i \otimes m_i \\ &= \sum_i t r_i (1 \otimes m_i) \end{aligned}$$

Thus, $T \otimes_R S$ is spanned by the $(1 \otimes m_i)$ and hence is module-finite.

4. Let $S = R[m_1, \ldots, m_n]$. Every element of $U^{-1}S$ is of the form,

$$\frac{s}{u} = \frac{1}{u} \cdot \frac{s}{1}$$

with $\frac{1}{u} \in U^{-1}R$ and since every $s \in S$ is of the form $s = \sum_i r_i m_i$ and then we have,

$$\frac{s}{1} = \sum_i \frac{r_i}{1} \cdot \frac{m_i}{1}$$

Thus, $U^{-1}S$ is spanned by $\frac{m_i}{1}$ as a $U^{-1}R$-module.

Problem 8. Comparing the Two Notions in Examples.
Decide whether each algebra is algebra-finite and whether it is module-finite over the indicated base ring. Justify your answers.

1. $\mathbb{K}[x]$ over $\mathbb{K}$.
2. $\mathbb{K}[x]/\langle x^2 \rangle$ over $\mathbb{K}$.
3. $\mathbb{Z}[i]$ over $\mathbb{Z}$.
4. $\mathbb{Z}[1/2]$ over $\mathbb{Z}$.
5. $\mathbb{K}[t]$ over $\mathbb{K}[t^2]$.
6. $\mathbb{K}[t, t^{-1}]$ over $\mathbb{K}[t]$.

Solution.

1. Algebra-finite not module-finite.
2. Algebra-finite and module-finite.
3. Algebra-finite and module-finite.
4. Algebra-finite not module-finite.
5. Algebra-finite and module-finite.
6. Algebra-finite not module-finite.

Let S be an R-algebra and let $s \in S$. We say that s is *integral over R* if there exists a monic polynomial with coefficients in R that s satisfies. In other words, s is integral over R if there are elements $a_1, \dots, a_n \in R$ such that

$$s^n + a_1 s^{n-1} + a_2 s^{n-2} + \cdots + a_{n-1} s + a_n = 0$$

in S, where each a_i acts through the structure map $R \to S$. The word *monic* is essential. For example, every rational number satisfies some nonzero polynomial over $\mathbb{Z}$, but only the integers satisfy monic polynomial equations over $\mathbb{Z}$. The monic condition is exactly what allows us to solve for high powers of s in terms of lower powers.

Problem 9. First Examples of Integral Elements.
Let S be an R-algebra.

1. Show that every element in the image of $R \to S$ is integral over R.
2. Show that every nilpotent element of S is integral over R.
3. Let $S = R[x]/\langle x^2 - r \rangle$ for some $r \in R$. Show that the image of x in S is integral over R.
4. Show that $i \in \mathbb{Z}[i]$ is integral over $\mathbb{Z}$.
5. Show that $t \in \mathbb{K}[t]$ is integral over $\mathbb{K}[t^2]$.
6. Show that $\frac{1}{2}(\sqrt{-3}+1) \in \mathbb{Q}(\sqrt{-3})$ is integral over $\mathbb{Z}[\sqrt{-3}]$.
7. Show directly that $1/2 \in \mathbb{Q}$ is not integral over $\mathbb{Z}$. Hint: Suppose $(1/2)^n + a_1(1/2)^{n-1} + \cdots + a_n = 0$ and clear denominators.

Solution.

1. Every element $s \in \operatorname{im}(\alpha)$ in the image of $\alpha : R \to S$ is of the form $\alpha(r) = s \implies s - \alpha(r) = 0$.
2. Assume $s^n = 0$. Then we have,
$$s^n + 0s^{n-1} + \cdots + 0 = 0$$
Thus, every nilpotent element of S is integral over R.
3. We have,
$$\overline{x}^2 + 0\overline{x} - r = 0$$
Thus, the image of x in S is integral over R.
4. We have,
$$i^2 + 0i + 1 = 0$$
Thus $i \in \mathbb{Z}[i]$ is integral over $\mathbb{Z}$.
5. We have,
$$(t)^2 + 0(t) - t^2 = 0$$
Thus $t \in \mathbb{K}[t]$ is integral over $\mathbb{K}[t^2]$.

6. We have,

$$\begin{aligned}\left(\frac{1}{2}(\sqrt{-3}+1)\right)^2 - \frac{1}{2}(\sqrt{-3}+1)+1 &= \frac{1}{4}\left(-2+2\sqrt{-3}\right) - \frac{1}{2}(\sqrt{-3}+1)+1 \\ &= \frac{1}{2}\left(\sqrt{-3}-1\right) - \frac{1}{2}(\sqrt{-3}+1)+1 \\ &= \frac{-2}{2}+1 = 0\end{aligned}$$

Thus $\frac{1}{2}(\sqrt{-3}+1) \in \mathbb{Q}(\sqrt{-3})$ is integral over $\mathbb{Z}[\sqrt{-3}]$.

7. Suppose for the sake of contradiction that $\frac{1}{2}$ was integral over $\mathbb{Z}$. Then for some n we would have,

$$\left(\frac{1}{2}\right)^n + a_{n-1}\left(\frac{1}{2}\right)^{n-1} + \cdots + a_1\left(\frac{1}{2}\right) + a_0 = 0$$

Then,

$$\begin{aligned} 0 &= 2^n\left(\left(\frac{1}{2}\right)^n + a_{n-1}\left(\frac{1}{2}\right)^{n-1} + \cdots + a_1\left(\frac{1}{2}\right) + a_0\right) \\ &= 1 + 2a_{n-1} + \cdots + 2^{n-1}a_1 + 2^n \\ \implies 1 &= 2(a_{n-1} + \cdots + 2^{n-2}a_1 + 2^{n-1})\end{aligned}$$

But this is a contradiction. Hence $\frac{1}{2}$ is not integral over $\mathbb{Z}$.

Problem 10. Integral Elements Give Finite Modules.
Let S be an R-algebra and let $s \in S$.

1. Suppose that the element $s \in S$ is integral over R, and
$$s^n + a_1 s^{n-1} + \cdots + a_n = 0$$
with $a_i \in R$. Show that every power s^m with $m \geq n$ is an R-linear combination of $1, s, \ldots, s^{n-1}$.
2. Deduce that $R[s]$ is generated as an R-module by $1, s, \ldots, s^{n-1}$.
3. Find a finite set of R-module generators for $S = \mathbb{K}[t]$ over $R = \mathbb{K}[t^2]$
4. Find a finite set of R-module generators for $S = \mathbb{Z}[i]$ over $R = \mathbb{Z}$.

Solution.

1. We will prove this by induction.

   Base Case: $m = n$.

   Then we have,
$$s^n = -(a_1 s^{n-1} + \cdots + a_n)$$
   which is a R-linear combination of $1, s, \ldots, s^{n-1}$.

   Inductive Hypothesis: Assume that s^{m-1} can be written as a R-linear combination of $1, s, \ldots, s^{n-1}$. Let $s^{m-1} = \sum_{i=0}^{n-1} r_i s^i$.

   Inductive Step: Then,
$$\begin{aligned} s^m = s \cdot s^{m-1} = s \cdot \sum_{i=0}^{n-1} r_i s^i \\ &= \sum_{i=1}^{n-1} r_{i-1} s^i + r_{n-1} s^n \\ &= \sum_{i=1}^{n-1} r_{i-1} s^i + r_{n-1} \left(-(a_1 s^{n-1} + \cdots + a_n) \right) \end{aligned}$$
   Thus every power s^m with $m \geq n$ is a R-linear combination of $1, s, \ldots, s^{n-1}$.

2. An element of $R[s]$ are finite sums of the form $\sum_i r_i s^i$ with $r_i \in R$. Since s^i with $i \geq n$ can be written as a R-linear combination of $1, s, \ldots, s^{n-1}$, $R[s]$ is spanned by $1, s, \ldots, s^{n-1}$ and hence is generated as a R-module by $1, s, \ldots, s^{n-1}$.

3. $\{1, t\}$
4. $\{1, i\}$

Problem 11. A Fundamental Criterion.
Let S be an R-algebra and let $s \in S$. Prove that the following are equivalent.

1. s is integral over R.
2. $R[s]$ is module-finite over R.
3. There exists a finitely generated R-submodule $M \subseteq S$ such that $1 \in M$ and $sM \subseteq M$.

Solution.

$(1 \implies 2)$

We have this from problem 10.

$(2 \implies 3)$

TODO:

$(3 \implies 1)$

Problem 12. Integral Elements Form a Subring.
Let $R \to S$ be an R-algebra. Let $\overline{R}^S$ denote the set of elements of S that are integral over R. When R is an integral domain and S is a field we call $\overline{R}^S$ the *integral closure of R in S*. The *integral closure* or *normalization* of a domain R is the integral closure of R in its field of fractions $S = \mathrm{Frac}(R)$. We denote the integral closure of R by $\overline{R}$.

1. If $x \in S$ is integral over R, prove that $R[x]$ is module-finite over R.
2. If $x, y \in S$ are integral over R, prove that $R[x, y]$ is module-finite over R. Hint: First show $R[x]$ is finite over R, then show $R[x, y]$ is finite over $R[x]$.
3. Let $s \in R[x, y]$. Show that multiplication by s defines an R-linear map
$$\mu_s : R[x, y] \longrightarrow R[x, y], \qquad f \longmapsto sf.$$
Use Cayley–Hamilton, or the determinant trick, to prove that s is integral over R.
4. Conclude that $x + y$, xy, and $-x$ are integral over R.
5. Conclude that $\overline{R}^S$ is a subring of S containing the image of R.

Solution.

1. We know this from Problem 10.

2. We know that if $x \in S$ is integral over R then, $R[x]$ is module finite over R. We also know that $y \in S$ is integral over R. That is there exist $a_i \in R$ such that,

$$y^n + a_1 y^{n-1} + \cdots + a_n = 0$$

Note that since $a_i \in R$ they are in $R[x]$ and this equation also holds in $R[x]$. That is, y is integral over $R[x]$, and $R[x][y]$ is module-finite over $R[x]$ and thus module-finite over R (Problem 7, part 2).

3. By Cayley-Hamilton we have the characteristic polynomial,

$$p(\mu_s) = \mu_s^n + p_1 \mu_s^{n-1} + \cdots + p_{n-1}\mu_s + p_n = 0$$

with $p_k \in R$. Applying μ_s to $1 \in R[x, y]$ gives us

$$p(s \cdot 1) = s^n + p_1 s^{n-1} + \cdots + p_{n-1} s + p_n = 0$$

which is a polynomial in s with coefficients in R and thus s is integral over R.

4. Then $x + y, xy, -x \in R[x, y]$, and multiplication by each of these is an R-linear map on the finitely generated R-module $R[x, y]$, so Cayley-Hamilton gives us a monic polynomial over R and thus $x + y, xy, -x$ are integral over R.

5. Thus we conclude that for $\alpha : R \to S$, every $\alpha(r)$ is integral over R and thus $\operatorname{im}(R) \subseteq S$. Moreover, part 4 shows that $\overline{R}^S$ is closed under addition, multiplication and inverses and thus is a subring of S.

Problem 13. **Computations with Integrality.**
Let $\mathbb{K}$ be a field.

1. Prove that the elements of $\mathbb{Q}$ integral over $\mathbb{Z}$ are exactly the integers. Hint: Write a rational number in lowest terms.
2. Show that $\mathbb{K}[t]$ is module-finite over $\mathbb{K}[t^2, t^3]$. Find explicit module generators.
3. Show that $\mathbb{K}[x, y]/\langle y^2 - x^3 \rangle$ is module-finite over $\mathbb{K}[x]$. Find explicit module generators.
4. Let $S = \mathbb{K}[x, x^{-1}]$, viewed as a $\mathbb{K}[x]$-algebra. Show that x^{-1} is not integral over $\mathbb{K}[x]$.
5. Deduce that $\mathbb{K}[x, x^{-1}]$ is algebra-finite but not module-finite over $\mathbb{K}[x]$.

Solution.

1. Assume for contradiction that some rational number $\frac{p}{q} \in \mathbb{Q}$, $q \neq 0$ is integral over $\mathbb{Z}$. Without loss of generality we assume $\gcd(p, q) = 1$. Then, there exists a polynomial with coefficients in $\mathbb{Z}$ as follows,
$$\left(\frac{p}{q}\right)^n + a_1 \left(\frac{p}{q}\right)^{n-1} + \cdots + a_n = 0$$
Then,
$$\begin{aligned} 0 &= q^n \left(\left(\frac{p}{q}\right)^n + a_1 \left(\frac{p}{q}\right)^{n-1} + \cdots + a_n \right) \\ &= p^n + a_1 q p^{n-1} + \cdots + a_n q^n \\ \implies p^n &= -q(a_1 p^{n-1} + \cdots + a_n q^{n-1}) \end{aligned}$$
That is $q \mid p^n$, but because $\gcd(p, q) = 1$ we have $\gcd(p^n, q) = 1$, so $q \mid 1 \implies q = \pm 1$. Thus, $\frac{p}{q} \in \mathbb{Z}$. Thus, the only elements of $\mathbb{Q}$ integral over $\mathbb{Z}$ are exactly the integers.
2. Since $\mathbb{K}[t]$ is module-finite over $\mathbb{K}[t^2]$ it is also module-finite over $\mathbb{K}[t^2, t^3]$. The module generators are $\{1, t\}$.
3. Since $\overline{y}^2 = \overline{x}^3$, every element of $\mathbb{K}[x, y]/\langle y^2 - x^3 \rangle$ can be written as $p(x) + q(x)y$ for $p, q \in \mathbb{K}[x]$, so $\{1, \overline{y}\}$ generates it as a $\mathbb{K}[x]$ module.

4. Assume x^{-1} is integral over $\mathbb{K}[x]$. Then, there exists a polynomial with coefficients in $\mathbb{K}[x]$ as follows,
$$\left(\frac{1}{x}\right)^n + a_1\left(\frac{1}{x}\right)^{n-1} + \cdots + a_n = 0$$
Then,
$$\begin{aligned} 0 &= x^n\left(\left(\frac{1}{x}\right)^n + a_1\left(\frac{1}{x}\right)^{n-1} + \cdots + a_n\right) \\ &= 1 + a_1 x + \cdots + a_n x^n \\ \implies 1 &= -x(a_1 + \cdots + a_n x^{n-1}) \end{aligned}$$
Hence a contradiction. Thus, x^{-1} is not integral over $\mathbb{K}[x]$.

5. Thus $\mathbb{K}[x, x^{-1}]$ is algebra-finite but not module-finite over $\mathbb{K}[x]$.

Problem 14. UFDs Are Integrally Closed.
Let R be a UFD with fraction field $S = \mathrm{Frac}(R)$.

1. Explain how we can think of S as an R-algebra?
2. Let $\alpha \in S$. Explain why we can write $\alpha = \frac{a}{b}$ with $a, b \in R$, $b \neq 0$, and a and b having no common irreducible factors.
3. Give an example of a domain R that is *not* a UFD for which part (2) is false.
4. Suppose $\alpha = a/b \in S$ is integral over R and
$$\left(\frac{a}{b}\right)^n + r_1 \left(\frac{a}{b}\right)^{n-1} + \cdots + r_{n-1} \left(\frac{a}{b}\right) + r_n = 0$$
for $r_1, \ldots, r_n \in R$. By clearing denominators show that b divides a^n.
5. Show that if a and b have no common irreducible factor and $b \mid a^n$, then b is a unit. Hint: Use unique factorization.
6. Conclude that $\alpha = a/b \in R$. Therefore every element of K that is integral over R already lies in R.
7. Conclude that every UFD is integrally closed.
8. Show that $R = \mathbb{K}[t^2, t^3]$ is not integrally closed by finding an explicitly integral element in $S = \mathrm{Frac}(R) = \mathbb{K}(t)$ not contained in R.
9. Show that $\mathbb{Z}[\sqrt{-3}]$ is not integrally closed. What does this mean about it being a UFD?

Solution.

1. We can think of S as an R-algebra with the natural inclusion
$$\iota : R \hookrightarrow S, \quad r \mapsto \frac{r}{1}$$
which is a ring homomorphism.
2. Since $S = \mathrm{Frac}(R)$, every element $\alpha \in S$ is by definition of the form a/b with $a \in R$ and $b \in R \setminus \{0\}$. Since R is a UFD we can cancel common irreducible factors of a and b and get a representation a/b with $\gcd(a, b) = 1$.
3. Let $R = \mathbb{Z}[\sqrt{-5}]$. Then, $\frac{6}{1} = \frac{2 \cdot 3}{1} = \frac{(1+\sqrt{5})(1-\sqrt{5})}{1}$ which fails unique factorization and the argument in part 2 breaks down.

4. We have,

$$
\begin{aligned}
0 &= \left(\frac{a}{b}\right)^n + r_1\left(\frac{a}{b}\right)^{n-1} + \cdots + r_{n-1}\left(\frac{a}{b}\right) + r_n \\
0 &= b^n\left(\left(\frac{a}{b}\right)^n + r_1\left(\frac{a}{b}\right)^{n-1} + \cdots + r_n\right) \\
0 &= a^n + r_1 b a^{n-1} + \cdots + r_n b^n \\
\implies a^n &= -b(r_1 a^{n-1} + \cdots + r_n b^{n-1}) \\
\implies b &\,|a^n
\end{aligned}
$$

5. If $\gcd(a,b) = 1$ then $\gcd(a^n, b) = 1$, then $b \mid a^n$ implies that b is a unit.

6. Let $bb' = 1$ then $\alpha = ab'/bb' = ab/1 \in R$. Thus, all integral elements of S over R are already in R.

7. Thus, $\overline{R}^S = R$ and R is integrally closed. Since R was an arbitrary UFD this holds for all UFDs.

8. $t \in S$ is integral over R with,

$$1(t^2) - 0(t) - t^2 = 0$$

where $t^2, 0, 1 \in R$. Thus R is not integrally closed.

9. $\mathbb{Z}[\sqrt{3}]$ is a UFD and by part 7 must be integrally closed.

An R-algebra S is called an *integral extension* of R if every element of S is integral over R. Said differently, the structure map $R \to S$ is integral if all elements of the target satisfy monic polynomial equations over the source. Integral extensions are the precise setting where algebra generation and module generation interact especially well. The exercises below prove the fundamental comparison theorem:

$$\boxed{\text{module-finite}} \quad \Longleftrightarrow \quad \boxed{\text{algebra-finite and integral}}.$$

More explicitly, the exercises that follow will guide us through the proof of the following theorem.

Theorem 0.3. *Let $R \to S$ be an R-algebra. The R-algebra S is module-finite over R if and only if it S is algebra-finite as an R-algebra.*

Note you will often, especially in algebraic geometry, see this expressed as finite is equivalent to finite type and integral. However, again we try to avoid this language for now.

Problem 15. Module-Finite Extensions Are Integral.
Let S be a module-finite R-algebra.

1. Fix $s \in S$. Show that multiplication by s defines an R-linear map
$$\mu_s : S \longrightarrow S \qquad \text{given by} \qquad x \longmapsto sx.$$
2. Apply the Cayley-Hamilton theorem (explain why you can apply it) to μ_s to deduce that s is integral over R.
3. Conclude that every module-finite R-algebra is integral over R.

Solution.

1. Let $S = R[m_1, \ldots, m_n]$. Then from S's ring axioms we have,
   - $\mu_s(x+y) = s(x+y) = sx + sy = \mu_s(x) + \mu_s(y)$
   - $\mu_s(rx) = s(rx) = r(sx) = r\mu_s(x)$

   Thus multiplication by s defines a R-linear map.
2. Since S is module-finite over R and μ_s is a R-linear map we can apply Cayley Hamilton with $I = R$. Then we get the characteristic polynomial,
$$p(\mu_s) = \mu_s^n + p_1\mu_s^{n-1} + \cdots + p_{n-1}\mu_s + p_n = 0$$
   with $p_k \in R$. Applying μ_s to $1 \in R$ gives us
$$p(s \cdot 1) = s^n + p_1 s^{n-1} + \cdots + p_{n-1}s + p_n = 0$$
   which is a polynomial in s with coefficients in R and thus s is integral over R.
3. Since s was an arbitrary element in S, this shows that all module-finite R-algebras are integral over R.

Problem 16. Finite Type Integral Extensions Are Module-Finite.
Suppose $S = R[s_1, \ldots, s_n]$ and each s_i is integral over R.

1. Prove that $R[s_1]$ is module-finite over R.
2. Prove that $R[s_1, s_2]$ is module-finite over $R[s_1]$, and hence module-finite over R.
3. Continue by induction to prove that S is module-finite over R.
4. Deduce that if S is algebra-finite and integral over R, then S is module-finite over R.

Solution.

1. We know this from Problem 10.
2. We know that if $s_1 \in S$ is integral over R then, $R[s_1]$ is module finite over R. We also know that $s_2 \in S$ is integral over R. That is there exist $a_i \in R$ such that,
$$s_2^n + a_1 s_2^{n-1} + \cdots + a_n = 0$$
Note that since $a_i \in R$ they are in $R[s_1]$ and this equation also holds in $R[s_1]$. That is, s_2 is integral over $R[s_1]$, and $R[s_1][s_2]$ is module-finite over $R[s_1]$ and thus module-finite over R (Problem 7, part 2).
3. Assume that $R[s_1, s_2, \ldots, s_{n-1}]$ is module-finite over $R[s_1, s_2, \ldots, s_{n-2}]$ and hence module-finite over R.

   Now consider, $R[s_1, s_2, \ldots, s_{n-1}][s_n]$. We know that $s_n \in S$ is integral over R. That is there exist $a_i \in R$ such that,
$$s_n^k + a_1 s_n^{k-1} + \cdots + a_k = 0$$
Note that since $a_i \in R$ they are in $R[s_1, s_2, \ldots, s_{n-1}]$ and this equation also holds in $R[s_1, s_2, \ldots, s_{n-1}]$. That is, s_n is integral over $R[s_1, s_2, \ldots, s_{n-1}]$, and $R[s_1, s_2, \ldots, s_{n-1}][s_n]$ is module-finite over $R[s_1, s_2, \ldots, s_{n-1}]$ and thus module-finite over R (Problem 7, part 2).
4. Since these were arbitrary rings, if S is algebra-finite and integral over R, then S is module-finite over R.

Problem 17. The Comparison Theorem.
Let S be an R-algebra. Prove that the following are equivalent.

1. S is module-finite over R.
2. S is algebra-finite over R and integral over R.

Then revisit Problem 9 and explain how the theorem classifies each example.

Solution.

We showed the direction $(2 \implies 1)$ in the previous question. So we only need to show $(1 \implies 2)$. Assume S is module-finite over R. From Problem 4 part 3 we know that S is algebra-finite over R and from Problem 15 part 3 we know that S is integral over R. Thus $1 \implies 2$.

1. $\mathbb{Z}[i]$ is algebra-finite and integral over $\mathbb{Z}$ and hence is module-finite.
2. $\mathbb{K}[t]$ is algebra finite and integral over $\mathbb{K}[t^2]$ and hence is module-finite.
3. $\mathbb{Q}(\sqrt{3})$ is not integral over $\mathbb{Z}[\sqrt{3}]$ and hence is not module-finite.
4. $\mathbb{Q}$ is not integral over $\mathbb{Z}$ and hence is not module-finite.

Problem 18. **Permanence of Integral Extensions.**
Let $R \to S \to T$ be maps of rings.

1. Prove that every quotient R/I is integral over R.
2. Suppose S is integral over R and T is integral over S. Prove that T is integral over R. Hint: If $t \in T$ satisfies a monic equation over S, first adjoin to R the finitely many coefficients appearing in that equation.
3. If S is integral over R and $J \subseteq S$ is an ideal, prove that S/J is integral over $R/(J \cap R)$.
4. If S is integral over R and $U \subset R$ is multiplicatively closed, prove that $U^{-1}S$ is integral over $U^{-1}R$.
5. More generally, if S is integral over R and $R \to R'$ is any ring map, prove that $R' \otimes_R S$ is integral over R'.

Solution.

1. We know that R is a R-module over itself. Then by Problem 7 part 1, R/I is module-finite and thus is integral.
2. Since T is integral over S, for all $t \in T$ we have,

$$t^n + s_1 t^{n-1} + \cdots + s_n = 0$$

where $s_1, \ldots, s_n \in S$. But S is integral over R so we have $R[s_1, \ldots, s_n]$ a module-finite extension over R. Moreover, t is integral over $R[s_1, \ldots, s_n]$ and $R[s_1, \ldots, s_n, t] = R[s_1, \ldots, s_n][t]$ is module-finite over $R[s_1, \ldots, s_n]$ and thus over R. Then by Problem 15, t is integral over R.

3. Since S is integral over R for any $s \in S$ we have,

$$s^n + r_1 s^{n-1} + \cdots + r_n = 0$$

Then for $\bar{s} \in S/J$ we have,

$$\bar{s}^n + \overline{r_1}\,\bar{s}^{n-1} + \cdots + \overline{r_n} = 0$$

But $\overline{r_i} = r_i + J \in S = r_i + (J \cap R) \in R$. Hence the coefficients $\overline{r_i}$ lie in $R/\mathfrak{p}$ and $\bar{s}$ is integral over $R/(J \cap R)$.

4. An element in $U^{-1}S$ is of the form $\frac{s}{u} = \frac{1}{u} \cdot \frac{s}{1}$. Since $\frac{1}{u} \in U^{-1}R$ it is integral over $U^{-1}R$. So, we only need to show that $\frac{s}{1}$ is integral over $U^{-1}R$. We know that s is integral over R and there exists some minimal monic polynomial of the form,

$$s^n + r_1 s^{n-1} + \cdots + r_n = 0$$

Then in $U^{-1}S$ by injectivity of η we have,

$$\left(\frac{s}{1}\right)^n + \left(\frac{r_1}{1}\right)\left(\frac{s}{1}\right)^{n-1} + \cdots + \left(\frac{r_n}{1}\right) = \left(\frac{0}{1}\right)$$

with $\frac{r_i}{1} \in U^{-1}R$.

5. We know that $R' \otimes S$ is generated by the tensors $1 \otimes s$. Since s is integral over R we have,

$$s^n + r_1 s^{n-1} + \cdots + r_n = 0$$

Applying the map $S \to R' \otimes_R S$, $s \mapsto 1 \otimes s$ gives us

$$(1 \otimes s)^n + r_1 (1 \otimes s)^{n-1} + \cdots + r_n = 0$$

Hence $1 \otimes s$ is integral over R' and since every element of $R' \otimes_R$ is a finite sum of products of these elements, the integrality is preserved.

Problem 19. A First Consequence for Maximal Ideals.
Let $A \subseteq B$ be an integral extension of domains.

1. Suppose B is a field. Prove that A is a field. Hint: If $0 \neq a \in A$, then $a^{-1} \in B$ is integral over A; write a monic equation for a^{-1} and multiply by a suitable power of a.

2. Let $R \to S$ be an integral ring map and let $\mathfrak{q} \in \operatorname{Spec}(S)$ with contraction $\mathfrak{p} = \mathfrak{q} \cap R$. Show that $S/\mathfrak{q}$ is integral over $R/\mathfrak{p}$.

3. Deduce that if $\mathfrak{q}$ is maximal, then $\mathfrak{p}$ is maximal.

4. Prove the converse: if $\mathfrak{p}$ is maximal, then $\mathfrak{q}$ is maximal. Hint: $S/\mathfrak{q}$ is an integral domain integral over the field $R/\mathfrak{p}$.

5. Conclude that under an integral ring map, a prime ideal upstairs is maximal if and only if its contraction downstairs is maximal.

Solution.

1. If $a \neq 0 \in A$ then $a^{-1} \in B$ is integral over A with
$$\left(\frac{1}{a}\right)^n + r_1\left(\frac{1}{a}\right)^{n-1} + \cdots + r_n = 0$$
And,
$$\begin{aligned} 0 &= a^n\left(\left(\frac{1}{a}\right)^n + r_1\left(\frac{1}{a}\right)^{n-1} + \cdots + r_n\right) \\ 0 &= 1 + r_1 a + \cdots + r_n a^n \\ \implies 1 &= a(r_1 + \cdots + r_n a^{n-1}) \end{aligned}$$
That is a has a inverse in A. Since a was an arbitrary element this implies that every non-zero element in A has a inverse and A is a field.

2. Since S is integral over R for any $s \in S$ we have,
$$s^n + r_1 s^{n-1} + \cdots + r_n = 0$$
Then for $\bar{s} \in S/\mathfrak{q}$ we have,
$$\bar{s}^n + \overline{r_1}\,\bar{s}^{n-1} + \cdots + \overline{r_n} = 0$$
But $\overline{r_i} = r_i + \mathfrak{q} \in S = r_i + \mathfrak{q} \cap R \in R = r_i + \mathfrak{p}$. Hence the coefficients $\overline{r_i}$ lie in $R/\mathfrak{p}$ and $\bar{s}$ is integral over $R/\mathfrak{p}$.

3. If $\mathfrak{q}$ is maximal, then $S/\,\mathfrak{q}$ is a field and then by part 1 $R/\,fp$ must be a field and hence $\mathfrak{p}$ must be maximal.

4. If $\mathfrak{p}$ is maximal, then $R/\,\mathfrak{p}$ is a field, and by part 2 $S/\,\mathfrak{q}$ is integral over $R/\,\mathfrak{p}$. Then for any $\bar{s} \neq 0 \in S/\,\mathfrak{q}$ we have the minimal monic polynomial,

$$\bar{s}^n + \overline{r_1}\,\bar{s}^{n-1} + \cdots + \overline{r_n} = 0$$

   And since $R/\,\mathfrak{p}$ is a field we have,

$$\begin{aligned} \overline{r_n} &= -\bar{s}(\bar{s}^{n-1} + \overline{r_1}\,\bar{s}^{n-2} + \cdots + \overline{r_{n-1}}) \\ 1 = \frac{\overline{r_n}}{\overline{r_n}} &= -\bar{s}\left(\frac{1}{\overline{r_n}}\bar{s}^{n-1} + \frac{\overline{r_1}}{\overline{r_n}}\,\bar{s}^{n-2} + \cdots + \frac{\overline{r_{n-1}}}{\overline{r_n}}\right) \end{aligned}$$

   Thus $\bar{s}$ has an inverse in $S/\,\mathfrak{q}$, and hence $S/\,\mathfrak{q}$ is a field and $\mathfrak{q}$ is maximal.

5. Thus under an integral ring map $\mathfrak{q}$ is maximal if and only if $\mathfrak{p}$ is maximal.

Problem 20. **More Examples of Integral Extensions.**
Decide whether each map is integral. If it is integral and algebra-finite, give module generators.

1. $\mathbb{Z} \to \mathbb{Z}[i]$.
2. $\mathbb{K}[t^2] \to \mathbb{K}[t]$.
3. $\mathbb{K}[t^2, t^3] \to \mathbb{K}[t]$.
4. $\mathbb{K}[x] \to \mathbb{K}[x, x^{-1}]$.
5. $\mathbb{K}[x] \to \mathbb{K}[x, y] / \langle y^2 - x^3 \rangle$.
6. $\mathbb{K}[x^2, xy, y^2] \to \mathbb{K}[x, y]$. Hint: Show that x and y are integral over the source.

Solution.

1. The map is integral and algebra-finite with $\{1, i\}$ as module generators.
2. The map is integral and algebra-finite with $\{1, t\}$ as module generators.
3. The map is integral and algebra-finite with $\{1, t\}$ as module generators.
4. The map is not integral.
5. The map is module-finite with generators $\{1, y\}$ and thus is integral.
6. The map is integral and algebra-finite with $\{1, x, y\}$ as module generators.

We now introduce a numerical invariant of a ring. A *chain of prime ideals of length d* in a ring R is a strictly increasing chain

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_d$$

of prime ideals. Thus the length counts the number of strict inclusions, not the number of prime ideals appearing in the chain. For a nonzero ring R, the *Krull dimension* of R is

$$\dim(R) := \sup\{d \in \mathbb{Z}_{\geq 0} \mid \text{there exists a chain } \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_d \text{ in } \operatorname{Spec}(R)\}.$$

If there are chains of arbitrarily large finite length, we write $\dim(R) = \infty$. The zero ring has no prime ideals; different authors use different conventions for its dimension, so we will avoid using the zero ring in basic dimension computations unless a convention is specified. We will eventually prove the following theorem, which will turn out to be key to computing the dimension of most rings.

Theorem 0.4. *If $\mathbb{K}$ is a field then $\dim \mathbb{K}[x_1, x_2, \ldots, x_n] = n$.*

Proving that the dimension of the polynomial ring $\mathbb{K}[x_1, x_2, \ldots, x_n]$ is at least n is relatively easy, all one must do is find an explicit chain of prime ideals of length n. You will do this in the exercises below. Proving the equality in the theorem requires us to prove the other inequality, i.e. we must show there is no chain of prime ideals of length more than n. This turns out to be surprisingly difficult. For the remainder of the worksheet you may assume this theorem unless it trivializes an exercise.

You will also prove that if R is any ring then $\dim R[x] \geq \dim R + 1$, again by exhibiting an explicit chain of prime ideals. Intuition tells us that this should be an equality, however, this turns out to be false unless we put additional hypotheses on R, namely that R be Noetherian. This highlights a somewhat tedious fact about dimension theory in commutative algebra: unless one assumes some form of finiteness condition (Noetherian, finitely generated $\mathbb{K}$-algebra, integral, etc.) there are many pathologies.

Using the correspondence between prime ideals and irreducible closed subsets of $\operatorname{Spec}(R)$, thee intuition is that longer chains of prime ideals correspond to longer chains of irreducible closed subsets in $\operatorname{Spec}(R)$, ordered in the opposite direction.

Problem 21. **Unpacking the Definition.**
Let R be a nonzero ring.

1. Show that $\dim(R) = 0$ if and only if every prime ideal of R is maximal.
2. Show that every field has Krull dimension 0.
3. Show that if R is an integral domain that is not a field, then $\dim(R) \geq 1$.
4. Let $I \subseteq R$ be an ideal. Use the correspondence between prime ideals of R/I and prime ideals of R containing I to describe $\dim(R/I)$ in terms of chains of prime ideals in R.
5. Prove that $\dim(R) = \dim(R_{\text{red}})$, where $R_{\text{red}} = R/\text{Nil}(R)$.

Solution.

1. ($\Longrightarrow$)

   Assume $\dim(R) = 0$. Then the longest chain of strict inclusions in any chain of prime ideals in R is zero. Thus every prime ideal must be maximal (otherwise there would be a maximal ideal containing it).

   ($\Longleftarrow$)

   Assume every prime ideal of R is maximal. Then the number of strict inclusions in any chain of prime ideals is zero, and thus $\dim(R) = 0$.

2. Since the only prime ideal in a field is $\langle 0 \rangle$, it is maximal and thus the field has Krull dimension 0.

3. If R is an integral domain $\langle 0 \rangle$ is a prime ideal and if R is not a field then $\langle 0 \rangle$ is not maximal so there must exist at least one maximal ideal containing it (since R is not the zero-ring). Hence, $\dim(R) \geq 1$.

4. $\dim(R/I) =$ length of longest chain of prime ideals in R containing I.

5. We know that $\text{Nil}(R) \subseteq \bigcap_i \mathfrak{p}_i$ where $\mathfrak{p}_i$ is a prime ideal in R. Then every chain of prime ideals in R contains $\text{Nil}(R)$ and by part 4,

$$\dim(R) = \dim(R_{\text{red}})$$

Problem 22. Dimension Zero Examples.
Compute the Krull dimension of each ring.

1. $\mathbb{K}$.
2. $\mathbb{K} \times \mathbb{K}$.
3. $\mathbb{K}[\epsilon]/\langle \epsilon^2 \rangle$.
4. $\mathbb{Z}/12\mathbb{Z}$.
5. $\mathbb{K}[x]/\langle x^n \rangle$ for $n \geq 1$.
6. $\mathbb{K}[x,y]/\langle x,y \rangle$.

Solution.

1. $\dim(\mathbb{K}) = 0$.
2. $\dim(\mathbb{K} \times \mathbb{K}) = 1$.
3. $\dim(\mathbb{K}[\epsilon]/\langle \epsilon^2 \rangle) = 0$.
4. $\dim(\mathbb{Z}/12\mathbb{Z}) = 0$.
5. $\dim(\mathbb{K}[x]/\langle x^n \rangle) = 0$
6. $\dim(\mathbb{K}[x,y]/\langle x,y \rangle) = 0$.

Problem 23. **Dimension One Examples.**

1. Prove that $\dim(\mathbb{Z}) = 1$.
2. More generally, prove that every principal ideal domain that is not a field has Krull dimension 1.
3. Prove that $\dim(\mathbb{K}[x]) = 1$.
4. Compute $\dim(\mathbb{K}[x]/\langle f \rangle)$ for a nonzero nonunit polynomial $f \in \mathbb{K}[x]$.
5. Compute $\dim(\mathbb{Z}[1/n])$ for an integer $n \geq 2$.

Solution.

1. We know that every non-zero ideal in $\mathbb{Z}$ is maximal (and the zero ideal is not maximal), so $\dim(\mathbb{Z}) = 1$.
2. We know from Math 71 that every non-zero prime ideal is maximal in a PID (and the zero ideal is not maximal), so every PID that is not a field has Krull dimension 1.
3. $\mathbb{K}[x]$ is a PID, so by part 2 $\dim(\mathbb{K}[x]) = 1$.
4. By Problem 21, part 4 we have that $\dim(\mathbb{K}[x]/\langle f \rangle) =$ length of longest chain of prime ideals in $\mathbb{K}[x]$ containing $\langle f \rangle$. But since f is non-zero and $\mathbb{K}[x]$ is a PID the longest chain containing $\langle f \rangle$ has length 1. Thus, $\dim(\mathbb{K}[x]/\langle f \rangle) = 1$.
5. Let $\mathbb{Z}[1/n] = U^{-1}\mathbb{Z}$, $U = \{1, n, n^2, \ldots\}$. Then prime ideals in $\mathbb{Z}[1/n]$ are prime ideals $\mathfrak{p}$ such that $\mathfrak{p} \cap U = \emptyset$. So $\langle 0 \rangle$ and any prime p such that $p \nmid n$ are prime ideals in $\mathbb{Z}[1/n]$. Thus, $\dim(\mathbb{Z}[1/n]) = 1$.

Problem 24. Building Longer Chains.
Let $\mathbb{K}$ be a field.

1. Prove that
$$\langle 0\rangle \subsetneq \langle x_1\rangle \subsetneq \langle x_1, x_2\rangle \subsetneq \cdots \subsetneq \langle x_1, \ldots, x_n\rangle$$
is a chain of prime ideals in $\mathbb{K}[x_1, \ldots, x_n]$.

2. Deduce that $\dim(\mathbb{K}[x_1, \ldots, x_n]) \geq n$.

3. Let R be any nonzero ring. Show that $\dim(R[x]) \geq \dim(R) + 1$ whenever $\dim(R)$ is finite. Hint: Extend a chain in R to $R[x]$, then add the ideal generated by the final prime together with x.

Solution.

1. First note that $x_i \notin \langle x_1, \ldots, x_{i-1}\rangle$ since $\langle x_1, \ldots, x_{i-1}\rangle$ is of the form $\sum_j k_j x_1^{j_1} \cdots x_{i-1}^{j_{i-1}}$ and x_i cannot be expressed as a combination of $x_1, \ldots, x_{i-1}$. Hence the containments are strict.

   Moreover the ideals are prime since for any ideal $\langle x_1, \ldots, x_{i-1}\rangle$ the quotient ring $\mathbb{K}[x_1, \ldots, x_n]/\langle x_1, \ldots, x_{i-1}\rangle = \mathbb{K}[x_i, \ldots, x_n]$ which is an integral domain.

   Hence,
$$\langle 0\rangle \subsetneq \langle x_1\rangle \subsetneq \langle x_1, x_2\rangle \subsetneq \cdots \subsetneq \langle x_1, \ldots, x_n\rangle$$
   is a chain of prime ideals in $\mathbb{K}[x_1, \ldots, x_n]$.

2. The chain in part 1 has length n so $\dim(\mathbb{K}[x_1, \ldots, x_n]) \geq n$.

3. Suppose $\dim(R) = n$. Then there exists a chain of length n in R,
$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$$
   We extend this to a chain in $R[x]$ by
$$\mathfrak{p}_0\, R[x] \subsetneq \mathfrak{p}_1\, R[x] \subsetneq \cdots \subsetneq \mathfrak{p}_n\, R[x] \subsetneq \mathfrak{p}_n\, R[x] + \langle x\rangle.$$
   Now note that Each $\mathfrak{p}_i\, R[x]$ is prime since the quotient $R[x]/\,\mathfrak{p}_i\, R[x] \cong (R/\,\mathfrak{p}_i)[x]$ is an integral domain since R/fp_i is. And $\mathfrak{p}_n\, R[x] + \langle x\rangle$ is prime since the quotient
$$R[x]/(\mathfrak{p}_n\, R[x] + \langle x\rangle) \cong R/\,\mathfrak{p}_n,$$
   which is an integral domain.

   Moreover the inclusions are strict. Suppose not, then if $\mathfrak{p}_{i-1}\, R[x] = \mathfrak{p}_i\, R[x]$ then contracting to R gives $\mathfrak{p}_{i-1} = \mathfrak{p}_i$, a contradiction. And the last inclusion is strict since $x \in \mathfrak{p}_n\, R[x] + \langle x\rangle$ but $x \notin \mathfrak{p}_n\, R[x]$ since elements of $\mathfrak{p}_n\, R[x]$ have all coefficients in $\mathfrak{p}_n$, and $1 \notin \mathfrak{p}_n$. Hence we have a chain of length $n+1$ in $R[x]$, so $\dim(R[x]) \geq \dim(R) + 1$.

Problem 25. Dimension and Quotients.
Let $\mathbb{K}$ be a field.

1. Compute $\dim(\mathbb{K}[x,y]/\langle xy\rangle)$. Hint: Every prime ideal containing xy contains x or y.
2. Compute $\dim(\mathbb{K}[x,y]/\langle x^2, xy\rangle)$. Hint: First pass to the reduced ring.
3. Compute $\dim(\mathbb{K}[x,y]/\langle y-x^2\rangle)$.
4. Find a chain of prime ideals of length 2 in $\mathbb{K}[x,y,z]/\langle xy\rangle$.
5. Assuming $\dim(\mathbb{K}[x_1,\ldots,x_n]) = n$, compute the dimension of $\mathbb{K}[x,y,z]/\langle z - y^2 + x^3\rangle$.

Solution.

1. We know that $\dim(\mathbb{K}[x,y]/\langle xy\rangle)$ = Longest chain containing $\langle xy\rangle$ in $\mathbb{K}[x,y]$.

   Every chain containing $\langle xy\rangle$ lies either above $\langle x\rangle$ or above $\langle y\rangle$, and by symmetry both give chains of length 2. So, $\dim(\mathbb{K}[x,y]/\langle xy\rangle) = 2$.

2. Let $R = \mathbb{K}[x,y]/\langle x^2, xy\rangle$. Then we have,

$$\dim(R) = \dim(R_{\text{red}}) = \dim(R/\langle\overline{x}\rangle) \cong \dim(\mathbb{K}[y]) = 1$$

3. We have,

$$\mathbb{K}[x,y]/\langle y-x^2\rangle \cong \mathbb{K}[x,x^2] = \mathbb{K}[x]$$

   Then,

$$\dim(\mathbb{K}[x]) = 1$$

4. A chain of prime ideals of length 2 is,

$$\langle\overline{x}\rangle \subsetneq \langle\overline{x},\overline{z}\rangle \subsetneq \langle\overline{x},\overline{y},\overline{z}\rangle$$

5. We have,

$$\mathbb{K}[x,y,z]/\langle z-y^2+x^3\rangle \cong \mathbb{K}[x,y,y^2-x^3] = \mathbb{K}[x,y]$$

   Then,

$$\dim(\mathbb{K}[x,y]) = 2$$

Problem 26. Local Dimension.
Let $\mathfrak{p} \in \operatorname{Spec}(R)$.

1. Recall that prime ideals of $R_{\mathfrak{p}}$ correspond to prime ideals $\mathfrak{q} \subseteq R$ with $\mathfrak{q} \subseteq \mathfrak{p}$. Use this to describe $\dim(R_{\mathfrak{p}})$ in terms of chains of prime ideals in R ending inside $\mathfrak{p}$.

2. Compute $\dim(\mathbb{K}[x]_{\langle x \rangle})$.

3. Compute $\dim(\mathbb{Z}_{\langle p \rangle})$ for a prime number p.

4. Find a chain showing that $\dim(\mathbb{K}[x, y]_{\langle x,y \rangle}) \geq 2$.

5. Assuming $\dim(\mathbb{K}[x, y]) = 2$, prove that $\dim(\mathbb{K}[x, y]_{\langle x,y \rangle}) = 2$.

Solution.

1. Since that prime ideals of $R_{\mathfrak{p}}$ correspond to prime ideals $\mathfrak{q} \subseteq R$ with $\mathfrak{q} \subseteq \mathfrak{p}$, $\dim(R_{\mathfrak{p}}) =$ Longest chain of prime ideals in R with $\mathfrak{p}$ as the maximal ideal.

2. $\dim(\mathbb{K}[x]_{\langle x \rangle}) = 1$.

3. $\dim(\mathbb{Z}_{\langle p \rangle}) = 1$.

4. $\langle 0 \rangle \subsetneq \langle x \rangle \subsetneq \langle x, y \rangle$. So $\dim(\mathbb{K}[x, y]_{\langle x,y \rangle}) \geq 2$.

5. If $\dim(\mathbb{K}[x, y]) = 2$ then the longest chain of prime ideals in $\mathbb{K}[x, y]$ is of length 2 so $\dim(\mathbb{K}[x, y]_{\langle x,y \rangle}) \leq 2$. Thus,

$$\dim(\mathbb{K}[x, y]_{\langle x,y \rangle}) = 2$$